Advanced Retrieval-Augmented Generation

A practical architecture for high-recall retrieval, trusted context assembly, and evidence-linked model outputs.

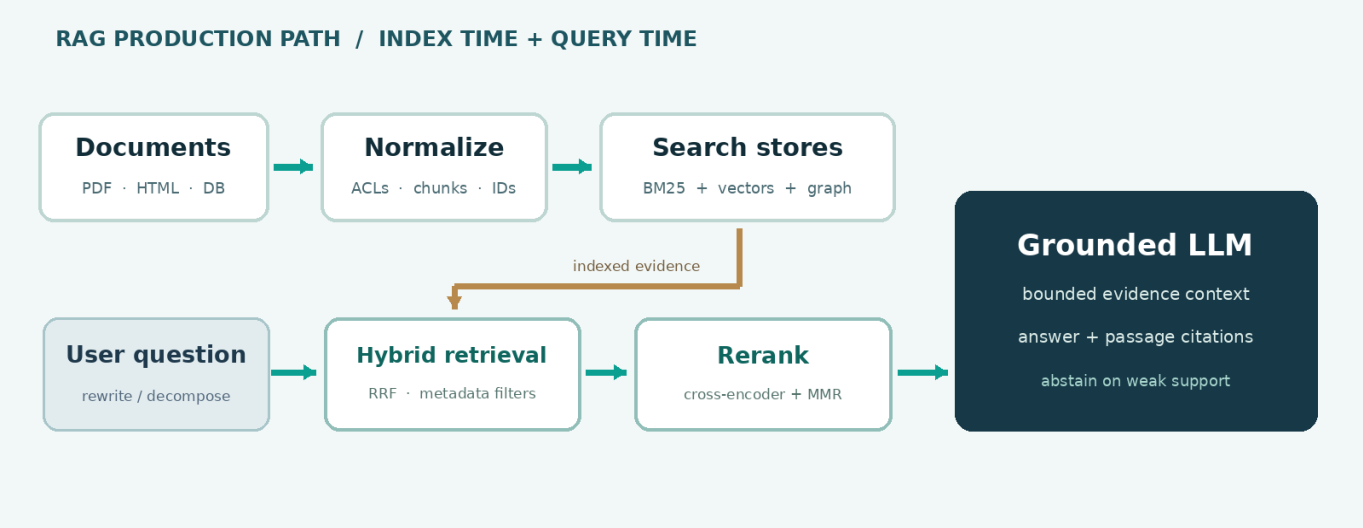


Figure 1. Separate index construction from live querying; retrieve broadly, then refine the evidence sent to the model.

1 The retrieval stack

Data plane. Parse PDFs, HTML, tables and records into traceable chunks with document IDs, version stamps, access-control tags and source offsets. Use semantic boundary-aware chunking; attach neighboring windows rather than indexing arbitrarily long passages.

Candidate generation. Combine lexical BM25 (exact IDs, rare terms) with dense embedding search (paraphrases). Graph or metadata filtering adds entity, relation and authorization constraints. Reciprocal rank fusion merges heterogeneous rankings without requiring comparable score scales.

2 Reasoning over evidence

Rerank + diversify. Apply a cross-encoder to a moderate candidate set, then enforce topical coverage and near-duplicate removal. Bound the final context by tokens and preserve passage-level identifiers; simply increasing top-k may dilute useful context.

Answer policy. Require claims to map to supporting excerpts, distinguish retrieval uncertainty from answer uncertainty, and abstain when relevant evidence is absent. Treat retrieved text as untrusted input: strip instruction-like content and preserve document-level access control.

Core fusion rule $\mathrm{RRF}(d) = \sum_{r \in R} 1 / (k + \mathrm{rank}_r(d))$. A larger k dampens rank differences; use a held-out relevance set to tune it.

Evaluation and design choices

TRADE-OFF MATRIX

Component	Technique	Strength	Principal trade-off
Candidate search	BM25 + embeddings	Exact term + semantic recall	Dual index upkeep
Rank fusion	RRF	Robust rank aggregation	Ignores raw score calibration
Reranking	Cross-encoder	Query / passage interaction	Adds inference latency
Global reasoning	GraphRAG communities	Corpus-level themes	Higher indexing cost
Long context	Selected evidence windows	Localized support + traceability	Context budget and truncation
Governance	ACL filters + provenance	Reduces leakage and audit risk	Requires strict source versioning

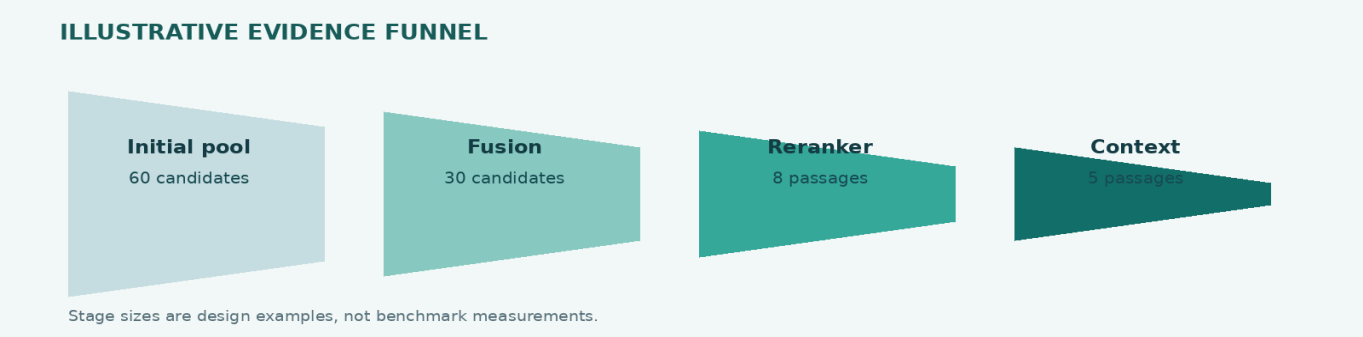


Figure 2. Illustrative candidate narrowing; counts are workflow examples, not empirical performance results.

Measurement framework

Retrieval: Recall@k tests whether gold evidence is surfaced; nDCG@k rewards relevant evidence near the top. Measure separate slices for rare entities, multilingual queries, recent records, and multi-hop reasoning.

Generation: check claim-level support, answer correctness, citation precision, and supported abstention. Score p50/p95 latency, retrieval-token footprint, and monetary cost along with quality.

Failure analysis and deployment

Typical failures: OCR corrupts evidence; chunk boundaries split facts; relevant documents are absent from the candidate pool; citations point to a source that does not entail the claim; stale indexes return superseded policy.

Release gates: maintain a human-reviewed evaluation set, test access isolation and prompt injection, monitor drift in search queries, and log the evidence IDs actually presented to the generator.

PRIMARY SOURCES

Lewis et al., RAG (2020) Microsoft GraphRAG docs TREC 2025 RAG